Concept Unlearning in Text-to-Image Diffusion Models

Status, findings, and a proposed change of direction

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28 October 2026

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Recommendation

Retire the method direction and write the structural result.

Across **164 erased models** and **1,215 attack measurements**, no re-implementation of any concept-erasure method survives relearning — including the four built specifically to resist it and both of our own. Three of four published methods are undone by a **single gradient step**. The concept returns under **benign fine-tuning with no adversary at all**, verified in the images.

One exception, found this week. ResAlign (NeurIPS 2025) — the authors' own released checkpoint — holds far longer than anything else. We reproduced their un-attacked figure exactly, verified it against a working control, and attacked it: it resists at 25 and 100 gradient steps where every other method falls at one to four, and breaks only by 300. It does not make erasure permanent; it makes the attack roughly **100x more expensive**.

That exception makes the paper stronger, not weaker. **Four families of method fail for one structural reason; a fifth escapes it, and we can say precisely what it does differently.**

What we need from you: a decision on this framing, and on how to spend the next ten days.

Results at a glance

All measured on SD-1.4, with pre-registered decision rules fixed before any number existed.

> **Scope, stated up front.** Except where marked *authors' checkpoint*, every arm is **our re-implementation** of the named method. Our own fidelity manifest records that nothing in this project is an official reproduction. The defensible sentence is *"across nine re-implementations, every arm is reversed"* — not *"every published method is reversed."*

1. It comes back with no adversary

Fine-tuning on **golf balls, french horns, cassette players and gas pumps** — no church prompt, no church image, no attacker. 200 steps, n=3, all cells valid.

method	suppression before	after benign fine-tuning	survives the control?	verified in images?
ESD	0.467	**1.000**	**yes — 0.80–0.93**	**yes — unambiguous churches**
IMMA *(built to resist fine-tuning)*	0.067	**0.533**	not tested	partly
UCE	0.000	0.444	**no — collapses to 0.000–0.200**	no — **claim withdrawn**

Utility rises in every cell and never falls. The model becomes better at everything, including the concept it was supposed to have forgotten — only possible if nothing was removed.

We tested this against ourselves twice. The obvious objection is that our downstream images came from the base model, so the fine-tune might simply drag the model back toward base behaviour. We re-ran it generating the downstream set from **the erased model itself** — nothing in training then comes from any model that can draw a church. **ESD survived unchanged. UCE collapsed to near zero**, and that half of the claim is withdrawn.

2. Four gradient steps

method	suppression before	1 step	2	4	8	16	steps to recovery
ESD	0.467	**0.933**	1.000	1.000	1.000	1.000	**1**
GIFT	0.467	**0.933**	1.000	1.000	1.000	1.000	**1**
IMMA	0.133	**0.800**	0.733	—	—	—	**1**
UCE	0.067	0.400	0.733	**0.867**	0.933	0.800	**4**

Complete at 75 of 75 cells, n=3 per point. **Three of the four published methods are undone by a single gradient step; the fourth takes four.** The better a method erases, the more steps it buys — and the most anything buys is four.

3. Every method, one number

method	built to resist fine-tuning?	recovery
UCE, ESD	no	1.000
GIFT, IMMA, FRP, Meta-Unlearning	**yes**	**1.000**
UTRAP, WSR-anchor — ours	**yes**	**1.000**
AdvUnlearn — authors' checkpoint	yes	**1.000** *(51 admissible cells)*

Our own method: **18 of 18 valid attacks at 1.000**, across three arms, one of which cleared the utility floor before being attacked. This is not an effort problem.

4. Attacker capacity does not help the defender

Complete threat ladder, 108 cells. L4 = rank-4 LoRA on attention projections, **0.09% of the UNet**.

arm	L4	L16	L32	full fine-tune
`uce_church`	**1.000** (9/9)	1.000 (6/9)	0.933 (6/9)	1.000 (17/17)
`esd_church`	**1.000** (9/9)	1.000 (6/9)	1.000 (6/9)	1.000 (22/24)
`gift_church`	**1.000** (9/9)	1.000 (6/9)	1.000 (6/9)	1.000 (15/15)
`imma_church`	0.800 (8/9)	**1.000** (6/9)	1.000 (6/9)	1.000 (14/15)

Elsewhere: a LoRA at **0.046%** of the UNet, and **768 scalars**, both reach full recovery.

5. The metric cannot tell erasure from damage

what the metric reported	what the image showed
suppression 0.40	a **clearly visible church** under speckle noise, scored `van_gogh=0.72`
suppression 0.0000	abstract texture mush — no church, no coherent alternative
leakage `church=0.546`	a **window-grid texture** containing no church

Suppression is a 12-way closed-set classifier with **no reject option**, so every image is forced onto a label. A robustness result measured on a broken model is meaningless.

6. The exception

	US rate	reference
base SD-1.4 *(positive control)*	**0.1817**	109 / 600 images
ResAlign higher-safety, un-attacked	**0.0000**	their reported figure, reproduced exactly
ResAlign after targeted attack, 25 steps	**0.0333**	all three learning rates valid

	US rate	reference
ResAlign after targeted attack, 100 steps	**0.0333**	4x the budget bought nothing
ResAlign after targeted attack, 300 steps	**0.1333**	it breaks — but ~100x later than anything else

The problem, and the bar

We set out to make a text-to-image diffusion model forget a concept in a way that survives someone fine-tuning it afterwards. Three requirements, in order:

1. the concept does not appear;
2. everything else still works;
3. both hold after an adversary retrains the weights.

The bar you set was explicit: *a clear, defensible advantage over existing methods. More computation is acceptable if the results are meaningfully better; comparable results are acceptable if substantially cheaper or simpler. An explicitly measured trade-off counts. Similar performance at similar or greater cost does not.*

You also wrote: *"You are explicitly authorized to substantially redesign, simplify, replace, or retire UTRAP if the evidence indicates that it cannot reach this objective within our resources."* **This report is that evidence.**

Requirements 1 and 2 are solved by published work and were never the difficulty. Requirement 3 is the open problem.

What was built

artifact	count
erased model checkpoints (arms)	164
attack measurements (cells)	1,215
generation evaluation panels	374
pre-registrations	~30
numbered findings	78

Nine published methods ported and pinned to upstream commits: ESD, UCE, GIFT, IMMA, FRP, FRAG, AdvUnlearn, Meta-Unlearning, UNDO. Two **authors' released checkpoints** obtained and attacked directly: AdvUnlearn and ResAlign. Five concepts (church, garbage truck, parachute, tench, Van Gogh) across three base models (SD-1.4, SD-1.5, BK-SDM-small).

Every experiment declared its success criterion, decision rule and noise floor in writing before any number existed.

Five families: how each works, what we tried, why it failed

We did not run any method once at default settings. Each was swept, and in two cases we repaired the published objective before testing it.

Family 1 — re-point the concept (edit methods)

method	how it works	what we swept	why it fails
UCE	closed-form least-squares edit of the cross-attention key/value matrices, re-mapping the concept's text embedding to a different output. No training	lambda across three orders of magnitude, preserve-scale, and the **embedding selection** — upstream edits one embedding, we edit a range, which is the only reason our arm erases at all	the UNet's visual machinery is **untouched by construction**. The attacker re-points the mapping back. Falls in **4 gradient steps**
ESD	fine-tunes the UNet so the concept's conditional score moves toward the negative-guidance direction	training steps, learning rate, three independent build seeds	the capability survives; only the score shifts. Falls in **1 gradient step**, and returns with **no attacker at all**

Family 2 — make the model hard to fine-tune (immunisation, meta-learning)

method	how it works	what we swept	why it fails
IMMA	bi-level: an inner loop simulates a fine-tuner, the outer loop makes that fine-tuning ineffective	alpha from 1e-5 to 1e-7, 300–1500 steps — finding a clean operating point at suppression 0.133, utility 0.942	rehearses against a **generic** fine-tuner. Falls in **1 gradient step** to one that targets the concept
GIFT	gradient-aware immunisation against malicious fine-tuning, with safe-concept retention	beta from 0.1 to 10.0, the full usable range	fully reversed at **every usable beta**, including under **its own DreamBooth threat model**

method	how it works	what we swept	why it fails
Meta-Unlearning	a three-term objective designed so that fine-tuning **on the erased concept** degrades the model — a self-destruct trap	we found variant A omits **two of Eq. 8's three terms**, restored them, then swept lambda; added common random numbers, fixed timesteps, micro-batched meta-gradients, Adam-epsilon throttling	the trap must fire on a gradient signal **benign data also produces**. It fires on normal use; at the one lambda leaving the model usable it buys **zero** robustness over its own ablation. Measured drift fell **below the random-walk floor**
UTRAP *(ours)*	train against unrolled inner relearning attacks so the path back is blocked	unroll depth K in {0, 1, 8, 32}, lambda sweeps, three arms	**18 of 18 attacks at 1.000**. The inner process differed from the real attacker on **seven axes**, so the hypothesis was never actually tested

Family 3 — delete the weights (pruning)

method	how it works	what we swept	why it fails
FRP	ranks weights by forget-importance minus retain-importance, zeroes the top rho%	rho, beta, lambda; plus a **mask-refill** experiment putting the mask back after the attack	**the mask is an initial condition, not a constraint.** Nothing holds the weights at zero, and the concept returned *through the weights FRP left alone*

Family 4 — destroy and rebuild (distillation)

method	how it works	what we swept	why it fails
UNDO	add noise to an erased model's weights, then distil the wreck back toward the erased teacher's outputs	alpha in {0.3, 0.5}, four damage-targeting modes, teacher swaps, budget sweeps	**erasure and utility turn out to be the same objective.** At alpha=0.3 it erased *perfectly* (0.0000 at four eval seeds) and halved utility to 0.517; at alpha=0.5, utility fell to 0.292
selective damage *(ours)*	weight the noise by retain-importance so damage spares what general generation needs	Fisher importance, its complement, and a **random control**	**it lost to the random control.** The importance signal the whole idea depends on is not there to find

Family 5 — make erasure a fixed point of the attack

method	how it works	result
ResAlign	penalises *how much the erasure degrades after simulated fine-tuning*, back-propagating through the inner loop via implicit hypergradients	**holds.** See the dedicated section below

Why four of five families fail — from first principles

There are exactly three ways an attacker can fail, and every method is one of them:

1. **The information is not there** — nothing remains from which the concept can be cheaply rebuilt.
2. **It is there but unreachable** — no short path through the loss landscape leads back.
3. **Reaching it destroys the model** — every path passes through ruin.

Route 1 fails because the concept is not separable. The information needed to draw a church is largely a *subset* of the information needed to draw the retained concepts — arches, towers, masonry, perspective. You cannot delete a subset while preserving the superset. Our damage-then-distil arm is the proof: perfect erasure at 0.000, utility halved to 0.517. The two objectives are the same objective.

Route 2 fails because the defender commits first. We fix the weights at release; the attacker then sees everything and chooses learning rate, optimiser, data and which parameters to touch. The defence must hold against every attack; the attack need work once. And fine-tuning toward the concept overlaps with fine-tuning toward legitimate tasks, so blocking one blocks the other.

Route 3 fails because the trigger is not separable either. A trap must fire on a gradient signal benign data also produces. Either it is too sensitive and fires during ordinary use — which we measured — or the attacker steps around it.

The compositional argument

A generative model does not store concepts; it composes them. A church is a reachable point in a space closed under composition, not an entry in a table. **You cannot remove the number seven from arithmetic while keeping three, four and addition.**

That is why the attacker needs only 25 steps and 768 scalars. The capability was never removed — only the binding between the word and the composition was moved, and a binding is cheap to restore.

What we can and cannot claim

We can claim, on 164 arms and 1,215 measurements, that **post-hoc weight-level concept erasure in these architectures does not survive relearning, that the attacker's cost is trivially small, and that the cause is the non-separability of the target from the retained set.**

We **cannot** claim unlearning is impossible in principle. Three escapes remain, and ResAlign occupies one of them.

The exception: ResAlign

ResAlign (NeurIPS 2025, arXiv:2507.16302) is the only published method in this area that demonstrates resistance to anything. We obtained **the authors' released checkpoints** — not a re-implementation — and tested them.

Their claim reproduces exactly

Their protocol: `unsafe.csv`, 3 images per prompt, 30 inference steps, their MHSC "sexual" detector (ViT-L-14/openai backbone plus their trained projection head), threshold 0.5.

arm	US rate	mean detector score
ResAlign higher-safety	**0.0000** (0/600)	0.0143
base SD-1.4 *(positive control)*	**0.1817** (109/600)	0.2053

Their reported un-attacked figure is 0.0000. **We measure 0.0000.**

The control is why this means anything. A detector reading zero on a defended model proves nothing unless it reads clearly non-zero on an undefended one. Base SD-1.4, through the identical pipeline, reads 0.1817 — 109 of 600 images flagged, mean score 14x higher. The instrument works and the zero is real.

It survives a targeted attack

Their paper tests *benign downstream fine-tuning*. We tested an adversary who wants the concept back and trains directly on it — the threat every other arm here was measured against. Attacker trains on base generations of 40 prompts; scored only on 60 prompts it never saw.

budget	learning rate	US after attack	% of base rate	attack valid?
25 steps	1e-5	**0.0000**	0%	yes
25 steps	5e-5	**0.0250**	14%	yes
25 steps	1e-4	**0.0333**	18%	yes
100 steps	1e-4	**0.0333**	18%	yes
300 steps	1e-4	**0.1333**	**73%**	yes

Every cell valid — probe loss fell, weights moved well past the fp32 floor. **This is a defence holding, not a failed attack.**

ResAlign is not an absolute defence, and that is the more useful result. At 25 steps — the budget that drives every other arm to full recovery — it reaches 0.0333 against a base rate of 0.1817, retaining about 82% of its protection. Quadrupling to 100 steps buys the attacker nothing. By 300 steps it breaks, reaching 73% of base.

So the quantity is a **cost**: ResAlign makes the attack roughly **100x more expensive** than against any other method, which fall at one to four steps. That is an explicitly measured trade-off — precisely the form the mandate accepts — rather than a claim of permanence.

What it does differently

Their outer objective, from their code, is one line:

```
loss = esd_loss(theta) + lambda - [esd_loss(theta*) - esd_loss(theta)] + w - retain_loss
```

where **theta*** is the model *after* a simulated fine-tuning run. In plain terms: *be erased now; don't become less erased after someone fine-tunes you; don't break everything else.*

The middle term is the entire contribution. They directly penalise **how much erasure degrades under fine-tuning**, and differentiate through the simulated fine-tuning with implicit hypergradients.

This is categorically different from a trap. Meta-Unlearning booby-traps the model: fine-tune on the concept and it self-destructs. That requires **detecting the attacker**, and the trigger signal is the same signal benign data produces — which is exactly the failure we measured. ResAlign never detects anything. It makes erasure a **fixed point** of fine-tuning.

Three further things they get right:

- **They harden against a distribution, not a point.** Fifteen inner configurations — learning rates {1e-6, 1e-5, 1e-4}, inner steps {5, 10, 20, 30}, optimizers {Adam, GD}, methods {LoRA, DreamBooth, standard} — sampled three at random per training step. IMMA and UTRAP each rehearsed against one.
- **They anneal the pressure.** lambda climbs 0.2 -> 0.8 on a sigmoid: erase first, then progressively weight post-fine-tuning survival.
- **Implicit hypergradients, not unrolling.** Fixed-point iteration lets them differentiate through 30 inner steps. Our UTRAP unrolled K in {0,1,8,32} and depth bought nothing.

One detail strengthens the result rather than weakening it. Their simulated attacker is LoRA rank-4, plus DreamBooth and standard fine-tuning. Our attack was a **full fine-tune of all 859M parameters** — out of distribution for what they trained against — and it still barely moved them. They generalised beyond their own training distribution.

The measurement problem

The second pillar is that the standard metric cannot distinguish erasure from damage, so positive results in this area are difficult to falsify as currently reported.

Suppression is scored as a 12-way closed-set CLIP accuracy over 15 images **with no reject option**. Every image is forced onto one of twelve labels. On 27 October we generated the images behind our own numbers and looked at them, and found all three of these:

1. the concept genuinely gone, something coherent in its place;
2. degenerate texture on the target prompt;
3. the concept **plainly present** but carrying an artifact the classifier misreads.

A result we had built on for days — an apparent robustness advantage — was a classifier being fooled by a noise overlay on a model that still drew churches perfectly. The companion utility metric cannot catch this either, because it is measured only on non-target prompts and never inspects the target image.

This also explains an inversion. Judged by the metric, our arm erased better than UCE. Judged by the images, UCE removes the church entirely and ours leaves it plainly visible. **The ranking reverses when you look.**

This criticism does not apply to ResAlign

Their metric is a **trained binary detector with a threshold**, not a closed-set argmax — and we verified its positive control. The defect we document does not touch their work, and **their evaluation is sounder than the one we have been criticising.** Any write-up must say so rather than sweep the field with one brush.

A related conflation

"Robust concept erasure" names two different threats, routinely confused:

tier	attacker may change	papers
1. Prompt / embedding	**nothing** — weights frozen	STEREO (CVPR 2025), AEGIS
2. Benign downstream fine-tuning	weights, on **unrelated** data	ResAlign
3. Targeted relearning	weights, on **the concept itself**	IMMA, GIFT, Meta-Unlearning claim this; our programme measures it

A method hardened at tier 1 can fall at tier 3 without its paper being wrong. STEREO's attacker cannot touch a single weight.

What we checked against ourselves

All 78 numbered findings were triaged against their own retraction banners. **Roughly a third of the headline claims in this project have been retracted or narrowed by subsequent work — almost always by us, against pre-declared falsifiers.** Five claims died on seed replication alone.

Four defects were identified as blocking a submission. Three are now closed:

	defect	status
D1	the evaluator used inconsistent label text — nine of twelve CLIP strings differ between scoring paths	**CLOSED.** Re-scored the same saved pixels under both label sets: target metric moves **0.0100** against a noise quantum of 0.067; utility **0.0121** against a floor of 0.042. Labels flip on 9.8% of images; **no conclusion changes**
D2	the validity gate's exclusion rate is non-monotone in pre-attack suppression (n=520)	**CLOSED.** Confirmed non-monotone — but the conclusion survives unfiltered (55/61 gated vs 58/74 ungated reach 1.000), and the only two arms that shift are **ours**, both looking *less* robust once the gate is removed. **The bias flatters defences, so our claim is understated**
D3	**nothing in this project is an official reproduction**	**OPEN, widened.** Our own fidelity manifest says so explicitly. Partially addressed: AdvUnlearn's and ResAlign's **authors' checkpoints** have now been attacked directly
D4	n=1 is pervasive — attacks are replicated at n=3 but every arm is a single build	**CLOSED on the key arm.** `esd_church` replicates across three build seeds: 0.4667 / 0.5333 / 0.4667, a full range of **one quantum**, utility flat at ~1.00

The contribution, and why it is publishable

The proposed paper:

> **Concept erasure in text-to-image diffusion is a rebinding, not a removal — four families of defence fail for one structural reason, a fifth escapes it, and the field's instruments cannot tell the difference.**

It is a diagnosis, not a tally. Negative results already exist for UCE and ESD, and the representation argument has been made for classifiers. What is missing is *why*. We give a mechanism, and the mechanism **predicts the numbers**: why 25 steps suffice, why 768 scalars suffice, why Fisher importance could not beat a random control, why damage-then-distil erases perfectly while halving utility.

It invalidates an instrument the field depends on. Not "these methods are weaker than reported" but "the standard measurement cannot distinguish erasure from damage" — supported by images rather than another accuracy number.

The coverage does not exist elsewhere. 164 arms, nine ported methods, two authors' checkpoints, six defence families, five concepts, three base models, 1,215 attack measurements — including the four methods built *specifically* to resist fine-tuning, which no one has put on a single bench under one attacker with honest cell accounting.

It answers a question another group has publicly posed. The AISTATS 2026 *Illusion of Unlearning* authors name extension to diffusion models as open future work.

The method failures are evidence, not embarrassment. Five directions built and closed, each by a rule written before it ran, **all failing for the same structural reason** — five independent probes of one obstruction. Our own method failing identically to the published ones is the strongest demonstration that this is not a matter of effort.

And it now contains a positive result. ResAlign works, we verified it rigorously, and we can state the mechanism that separates it from everything that fails.

Evidence behind each claim

claim	measurement
erasure fails universally	86 concept-matched arms: **56** fully recovered, 21 void on a harness defect, 6 destroyed pre-attack, 3 never erased. No arm unaccounted for
methods built to resist also fail	GIFT **12/12 valid attacks to 1.000**, including under its own DreamBooth threat model - IMMA **1.000 over 14 of 15 cells** - AdvUnlearn's **authors' checkpoint, 51 admissible cells, 1.000**
our own method fails identically	UTRAP: **18 of 18 valid attacks at 1.000**
the attacker is trivially cheap	**<=100 steps across six families at n=3** - **25 steps, zero variance** - LoRA at **0.046%** of the UNet - **768 scalars** - rank-4 LoRA reverses UCE over **9 of 9 cells**

claim	measurement
the concept is not separable	Fisher-weighted damage **lost to a random control** - damage-then-distil erased perfectly (0.0000, four seeds) while utility halved to **0.517** - FRP's pruned weights were never relearned
not an artifact of one setting	three base models including a half-size pruned architecture - five concepts - held-out prompts
apparent robustness can be the attacker handicapping itself	DreamBooth "resistance" vanished at **prior-preservation weight 0**
the metric cannot distinguish erasure from damage	a visible church scored **`van_gogh=0.72`** - a window grid scored **`church=0.546`**
the estimator itself is suspect	the gate's exclusion rate is **non-monotone in `T_before` (n=520)** — found in our own pipeline
the process is falsifiable	~30 pre-registrations - **roughly a third of our own headline claims retracted** - five died on seed replication

Ten days to a submittable contribution

days	work	why it matters
1–2	Finish the ResAlign budget sweep (300, 1000 steps, running); repeat on the higher-utility checkpoint; image-audit the outcome	Establishes **where** the one working defence breaks. A method that buys 40x the attacker budget is a different claim from one that holds indefinitely
3–4	Attack every author-released checkpoint obtainable; correct attribution throughout	Converts *"our re-implementations fail"* into *"the authors' own models fail."* AdvUnlearn and ResAlign are already done
5–6	Representation probe: can a classifier recover the concept from an erased model's internal activations?	Upgrades the mechanism claim from *we refuted every removal hypothesis* to *here is where the concept still lives*
7–8	Assemble figures and tables; raise any remaining headline to n=3	Four results already carry n=3; the load-bearing arm replicates across build seeds
9–10	Write	—

Already finished, needing no further compute: the 164-arm reversal result, the complete threat ladder, the cost curve, the benign-recovery result with its control and image verification, the ResAlign reproduction and targeted attack, and three of the four blocking defects.

The decision we need from you

1. **Commit to this framing?** The paper is now *"four families fail for one reason, a fifth escapes it"* rather than *"nothing works."*
2. **Target venue and deadline** — this sets how much of the fidelity work in days 3–4 is worth doing.
3. **Does the NSFW evaluation line continue?** ResAlign's concept is sexual content; the work is on their own benchmark with their own detector.
4. **Is the representation probe worth two of the ten days**, or do we spend them writing?

References

Methods ported and measured

work	venue / id	our use	pin
UCE — Unified Concept Editing, Gandikota et al.	arXiv:2308.14761	our re-implementation; embedding selection deviates deliberately from upstream and is why our arm erases	`7c724d9`
ESD — Erased Stable Diffusion, Gandikota et al.	—	our implementation; carries the benign-recovery headline	—
GIFT	2025	our implementation; **no official code exists**	—
IMMA	2024/25	our implementation; scope is under half the reference's	`2f55041`
Meta-Unlearning on Diffusion Models	**ICCV 2025**, arXiv:2410.12777	ported and measured; also establishes that T2I erasure is reversed by malicious fine-tuning	`5a2a760`
AdvUnlearn, OPTML-Group	—	**authors' released checkpoint**, attacked directly — 51 admissible cells, fully reversed	`bca3b279`
ResAlign	**NeurIPS 2025**, arXiv:2507.16302	**authors' released checkpoints**, verified and attacked	`14775d3`
UNDO — Distillation Robustifies Unlearning	**NeurIPS 2025**, arXiv:2506.06278	ported to T2I; the damage-then-distil direction, closed	—
FRAG / FRP	—	our port; no released code	`7fa3b5e`

Prior work bounding our claim

work	venue / id	relevance
An Illusion of Unlearning? — Gao, Unal, Rangamani, Zhu	**AISTATS 2026**, arXiv:2604.08271	the same argument for **classifiers**; **names extension to diffusion as open future work**
Erased or Dormant?	arXiv:2505.16174	publishes the reversibility finding — **for UCE and ESD only**
STEREO — Srivatsan et al.	**CVPR 2025**	"robust concept erasure" meaning robust to adversarial **prompts**, weights frozen
AEGIS	arXiv:2602.06771	also prompt-space robustness
PGU	2026, MBZUAI thesis	post-hoc hardening; constrains updates orthogonal to a retain subspace
From Dormant to Deleted	arXiv:2505.22310	weight-distance predictor; our adaptation inverted its direction

Caveat. The literature sweep of 28 October read **abstracts and metadata only** — the session's web-search budget was exhausted and the arXiv API returned HTTP 429 throughout. Only ResAlign, Meta-Unlearning, STEREO and the AISTATS paper were read directly. Characterisations of the others must be checked against full texts before submission.